

人工智能学院课程——现代控制理论

自抗扰控制的思想与基本结构

薛文超

中国科学院数学与系统科学研究院

2024.12

wenchaoxue@amss.ac.cn

概要

- 自抗扰控制的思想
 - 自抗扰控制的非线性设计方法
 - 自抗扰控制的线性设计方法
 - 自抗扰控制与PID控制的关系
 - 自抗扰控制的一些应用实例
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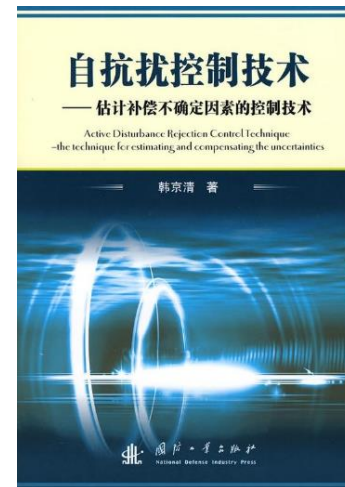
自抗扰控制的提出者

韩京清：1937-2008

- 1960s: 线性最速控制理论中的“等时区”方法
 - 1970s: 最优控制用于拦截问题中的导引律设计
 - 1980s: 线性系统理论的构造性方法、中国控制系统计算机辅助设计工程化软件系统(CADCSC)
 - 1995-2008: 自抗扰控制
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韩京清：1937-2008



- 宋健, 韩京清. 线性最速控制系统的分析与综合理论. 数学进展, 1962, 5(4): 264-282.
 - 韩京清. 拦截问题中的导引律. 北京: 国防工业出版社, 1977.
 - 韩京清. 控制理论-模型论还是控制论. 系统科学与数学, 1989, 9(4): 328-335.
 - 韩京清. 一类不确定对象的扩张状态观测器. 控制与决策, 1995, 1: 85-88.
 - 韩京清. 自抗扰控制器及其应用. 控制与决策, 1998, 1: 18-23.
 - 韩京清. 自抗扰控制技术. 北京: 国防工业出版社, 2008.
 - J. Han. From PID to Active Disturbance Rejection Control. IEEE Trans. Ind. Electron. , 2009, 56(3): 900-906.
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控制设计中的不变性原理(invariant principle), 1950s-60s

在以后,所謂不变性,我們將理解为扰动补偿,即所研究的自动調节系統(包括測量系統和仪表)能达到与作用在其上的扰动完全无关或部分无关。

扰动調节原理发展的历史道路是比較长的,它的产生还在上世紀中,毫無疑問是自动調节中的最古老原理。在我們这一世紀前的几千年,由阿拉伯人創造的第一批风力磨的控制裝置,已經包含了按負載来控制的元件。在这种风力磨中,磨子的轉速是按外部負載力矩的作用来改变风帆而进行調节的。

扰动控制和調节的現代景况,往往与法国学者龐雪萊^[23]和俄国学者契柯列夫的名字有关,他們給出了按被控系統外部負載作用而工作的調节器的描述和技术設計。历史上之所以注意龐雪萊裝置(1830年)和契柯列夫裝置(1847年)是由于按偏差原則构成的調节器(瓦特調节器)在应用方面沒有成效(在19世紀中叶),这些調节器一时还不很完善。在此时,出現了在原則上不同于瓦特調节器,并直接根据所測負載变化来进行調节的新型調节器,这对整个自动控制 and 自动調节产生了重要的影响。后来,理論証实了將按偏差原則(瓦特原理)的調节与按負載的調节原則(龐雪萊-契柯列夫原理)相結合,在許多情況下是最好的解决办法。

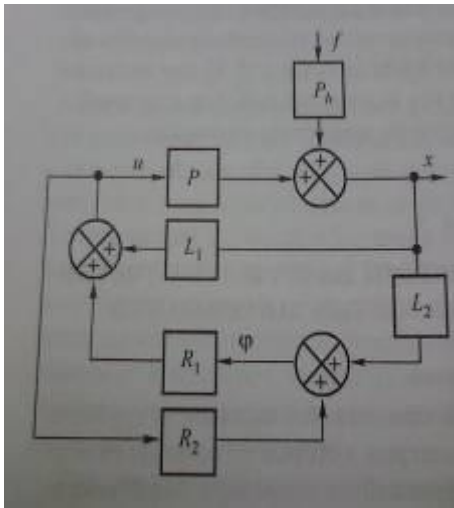
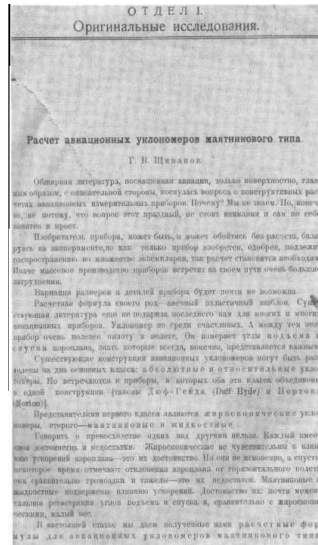
扰动調节

——扰动补偿和不变性——

〔苏联〕Г. М. 烏兰諾夫 著

胡保生 譯 徐俊榮 校

Grigoriy Shipanov (1903-1953)



- Soviet Engineer, Professor.
- Problem of automatic control: making output “invariant” in the presence of disturbance.
- 1938, dissertation, not understood by the university committee members: 6 voted “No,” 4 voted “Yes,” 20 didn’t vote.
- The same committee awarded Shipanov the title of Professor.
- 1938, hired by Kulebyakin, head of the Institute of Automatics and Telemechanics, finished his research there.
- 1939, a short publication of his completed research provoked heated debate on the ideological basis.
- 1941, publicly accused of creating a “fantastic” controller by mathematical speculations instead of meeting demands of a growing Soviet economy.

ENGINEERING CYBERNETICS

H. S. TSIEN

*Daniel and Florence Guggenheim Jet Propulsion Center
California Institute of Technology
Pasadena, California*

内外扰动的补偿问题

■ Ataraxia (disturbance free)

Ataraxia: 平静, 冷静; 心平气和; 心神安定之意, 词性为名词 (n.), 词根来自古希腊的一个哲学流派观点

15.6 Design for Stable Operation. Stability of any control system means that the design performance of the system will be obtained even with the presence of internal and external disturbances. We have seen how this requirement is satisfied in the case of conventional servo-mechanisms and other more general control systems in the preceding chapters. For optimizing control systems, the essential part of the operation is the proper coordination of the input drive with the output behavior, so that the output stays within a close neighborhood of the optimum. This operation must not be influenced by internal and external disturbances. When this is achieved by a good design of the system, we have stable operation.

最优控制中的扰动补偿问题

Accommodation of disturbances in optimal control problems†

C. D. JOHNSON

University of Alabama in Huntsville, Huntsville, Alabama 35807

[Received in revised form 19 May 1971]

Modern theories of automatic control have virtually ignored one of the oldest and most fundamental problems of control—the problem of controlling systems subjected to persistently acting, unknown external disturbances.

In this paper the problem of control in the face of unknown disturbances is studied from the optimization point of view. It is first shown that a straightforward approach to the problem leads to control laws that are hopelessly unrealizable. A novel alternative approach is then proposed which leads to physically realizable control laws which perform optimally in the face of a broad class of unknown external disturbances. The proposed method is illustrated by posing and solving a modified version of the classical linear-quadratic regulator problem, in which persistently acting disturbances have been added.

伺服系统中外扰补偿问题

Accommodation of External Disturbances in Linear Regulator and Servomechanism Problems

C. D. JOHNSON, MEMBER, IEEE

III. AN EFFECTIVE MATHEMATICAL MODEL

OF DISTURBANCES

Disturbances are by definition plant inputs $w(t) = (w_1(t), \dots, w_p(t))$ which cannot be manipulated by the designer and are not completely known beforehand. Various schemes have been proposed for mathematically modeling partially unknown functions $w(t)$ of this type. The traditional approach consists of using classical Fourier series methods on representative experimental recordings of $w(t)$ to estimate the discrete harmonic content of the disturbance. Another method consists of treating the disturbance as a random process and experimentally computing the mean, covariance, power spectral density, and other statistical properties of $w(t)$. This method is frequently used to model disturbances $w(t)$ as a continuous fusillade of various hues of colored noise. A less common, and conceptually different, way to model a partially unknown function $w(t)$ is to give a differential equation which $w(t)$ is known to satisfy. This latter technique, called disturbance "state" modeling, is based on the idea of characterizing the possible waveform modes of disturbances and will be used exclusively in this paper. We will show that this ap-

two parts:

$$u = u_c + u_R \quad (4)$$

where the component u_c is assigned the task of counteracting the disturbance $w(t)$ and the component u_R is responsible for regulating the state $x(t)$. In particular, u_R should choose the counteraction component of $u(t)$ as

$$u_c(t) = -\Gamma(t)\hat{w}(t) \quad (8)$$

and the regulating component of $u(t)$ as

$$u_R(t) = K(t)\hat{x} \quad (9)$$

where $\hat{w}(t)$, $\hat{x}(t)$ are accurate estimates of $w(t)$, $x(t)$ which we will obtain from $y(t)$ by an on-line real-time state reconstructor device, and $K(t)x(t)$ is the control which would

$$\begin{aligned} w(t) &= H(t)z; & w &= (w_1, \dots, w_p) \\ \dot{z} &= D(t)z + \delta(t); & z &= (z_1, \dots, z_p), \end{aligned}$$

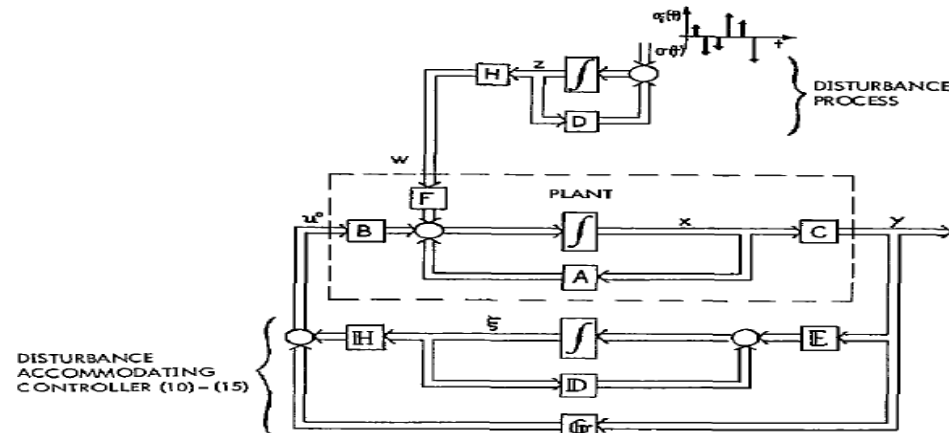


Fig. 1. Optimal linear regulator with disturbance accommodating controller (counteraction mode).

具有未知输入系统的观测器

Observers for systems with unknown and inaccessible inputs†

J. S. MEDITCH‡ and G. H. HOSTETTER§

(Observer theory for constant-coefficient linear systems is extended to systems with unknown inaccessible inputs. Two types of observers for such systems are defined, and simplified criteria for their existence are developed. A theorem regarding the structure of these observers is given which allows the application of results for systems with single unknown inputs to systems with multiple unknown inputs.

Observer theory (Luenberger 1964, 1966, 1971, Anderson and Moore 1971) generally requires that all inputs to the system whose state is being estimated be known. In practice, there exist many situations in which the system inputs are unknown (even in a statistical sense) and inaccessible (i.e. they cannot be measured). These typically take the form of internal and/or external disturbances in the plant, and unknown errors in the output measurements.

vable. Davison (1970, 1972) and Johnson (1968, 1970) have obtained some results for feedback controllers in this case, but not from the standpoint of observer theory. The relation of their results to observer theory has been commented on elsewhere (Hostetter and Meditch 1973 a).

系 统 科 学 与 数 学

J. Sys. Sci. & Math. Scis.

9 (4) (1989), 328—335

控制理论——模型论还是控制论¹⁾

韩 京 清

(中国科学院系统科学研究所)

Journal of System Sciences and Mathematical
Sciences, 9(4) (1989), 328-325

Control Theory: the Doctrine of Model or the Doctrine of Control?¹

Jing Qing Han

(Institute of System Sciences, Chinese Academy of Sciences)

(Translated by Zhiqiang Gao, Cleveland State University, Feb. 19th, 2013)

控制理论：控制论还是模型论？

在导引理论中，控制律是依靠对象实时状态的某些信息来建立的。她不大去考虑对象数学模型的细节，不把系统分成线性的和非线性的而采用不同的研究方法。根据视线信息有“三点法”，根据位置信息有“平行接近法”，根据视线和视线角速信息有“比例导引”，根据位置和速度信息有“前置量法”、“预测导引法”等。在作理论分析时，常常采用最简单的线性控制系统模型：

$$\begin{cases} \dot{x} = v, \\ \dot{v} = a(t) + u(t), \end{cases} \quad (2.1)$$

其中， $u(t)$ 是控制量， $a(t)$ 是所谓的“固有加速度”，是许多非线性效应的随时变化量。在这里，通常不把 $a(t)$ 写成状态变量的非线性函数。有时为了提高精度，对 $a(t)$ 作出适当估计是必要的，但不一定要知道 a 与状态变量之间非线性函数关系本身。在这里，并不大关心非线性对象的全局形态，关心的焦点是如何根据一个过程提供的实时信息来搞好这个过程的控制问题。这是导引理论提供给我们的主要思想方法。

根据系统对信号的某些响应特征或过程的某些实时信息来确定控制好一个过程的控制律，是与靠系统的数学模型去找控制律的方法完全不同的思考方式。这种思考方式，相对于模型论方法，可以称为控制理论中的“控制论”方法。

线性和非线性系统的标准型(Canonical form)

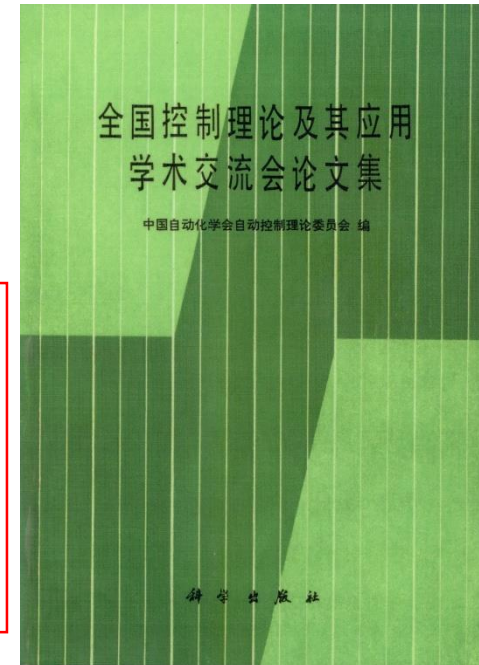
线性系统的结构与反馈系统计算

韩京清
(科学院数学所)

自抗扰控制源头：韩京清，第1届中国控制会议，1979年

The systems, linear or nonlinear, under some conditions, can be transformed into the **canonical form of cascade integrators via feedback.**

-Han J. The structure of linear system and computation of feedback system, in: Proc. National Conference on Control Theory and Its Application, Academic Press, 1980



对应于这个阵的系统叫做积分器串联系统。这是系统中最简单的系统。任意完全能控系统经适当坐标变换和状态反馈均可化成这种形式。由于反馈不改变系统的能控性，因此研究与系统的能控性有关问题，都可用对应的积分器串联形式来讨论。这种现象不仅对线性系统存在，而且在一类非线性控制系统中也成立。

积分串联型：线性 and 非线性系统的标准型

The systems, linear or nonlinear, under some conditions can be transformed into the **canonical form of cascade integrators via feedback.**

$$\begin{aligned}\dot{x} &= Ax + Bu, \\ y &= Cx\end{aligned}$$

$$A = \begin{bmatrix} 0 & 1 & \cdots & 0 & 0 \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ & & & & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

纯积分串联型系统：线性/非线性系统的标准型

积分串联型的获得：通过微分几何精确线性化系统

1980s-1990s: 非线性系统的理论分析是热点

- R. W. Brockett, **Feedback invariants for nonlinear systems**, 1978 IFAC.
- B. Jakubczyk and W. Respondek, **On linearization of control systems**, 1980.
- B. Charlet, J. Levine, **On dynamic feedback linearization**, 1989.

differential geometry

- M. Fliss, **Generalized controller canonical forms for linear and nonlinear dynamics**, 1990.

differential algebra

通过微分几何的工具精确线性化系统

Isidori A., Nonlinear Control System, Springer-Verlag

积分串联型的获得：通过微分几何精确线性化系统

当假设系统的非线性函数光滑并且精确已知时：

SISO nonlinear system $\dot{x} = f(x) + b(x)u, x \in R^n$
 $y = h(x)$

如果系统相对阶为n, 则存在一个等价变换 $z = T(x)$ 使得系统变化为如下系统

$$\begin{aligned} \dot{z} &= Az + B \frac{1}{\beta(x)} (u - \alpha(x)), \\ y &= Cz \end{aligned} \quad A = \begin{bmatrix} 0 & 1 & \cdots & 0 & 0 \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ & & & & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, C = [1 \ 0 \ \cdots \ 0 \ 0],$$
$$\alpha(x) = -\frac{L_f^n h(x)}{L_b L_f^{n-1} h(x)}, \beta(x) = \frac{1}{L_b L_f^{n-1} h(x)}$$

则设计状态反馈控制律 $u = \alpha(x) + \beta(x)v$ 得到纯积分串联型系统

$$\begin{aligned} \dot{z} &= Az + Bv, \\ y &= Cz \end{aligned}$$

问题：系统具有不确定性时，如何使系统变为纯积分串联型？

扩张状态观测器：估计纯积分串联型之外的总扰动

The Extended State Observer for A Class of Uncertain Systems, Control and Decision, 1995, 10(1): 85-88.

设有受到未知外扰作用的非线性不确定对象

$$\begin{cases} \dot{x}_1 = x_2 \\ \dots \\ \dot{x}_{n-1} = x_n \\ \dot{x}_n = f(x_1, x_1, \dots, x_n, w(t), t) + bu \end{cases}$$

其中 $f(x_1, x_1, \dots, x_n, w(t), t)$ 为未知函数， $w(t)$ 为未知干扰。若 $x_1(t)$ 为量测量，那么能否构造出不依赖于 $f(\cdot)$ 的非线性系统，使得它能由量测量 $x_1(t)$ 估计出被扩张的系统状态变量呢？

扩张状态观测器：估计纯积分串联型之外的总扰动

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状态空间的形式

$$\begin{cases} \dot{X} = AX + B(f(\cdot) + u), \\ y = CX \end{cases}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}, A = \begin{bmatrix} 0 & 1 & \dots & 0 & 0 \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ & & & & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix},$$
$$C = [1 \quad 0 \quad \dots \quad 0 \quad 0]$$

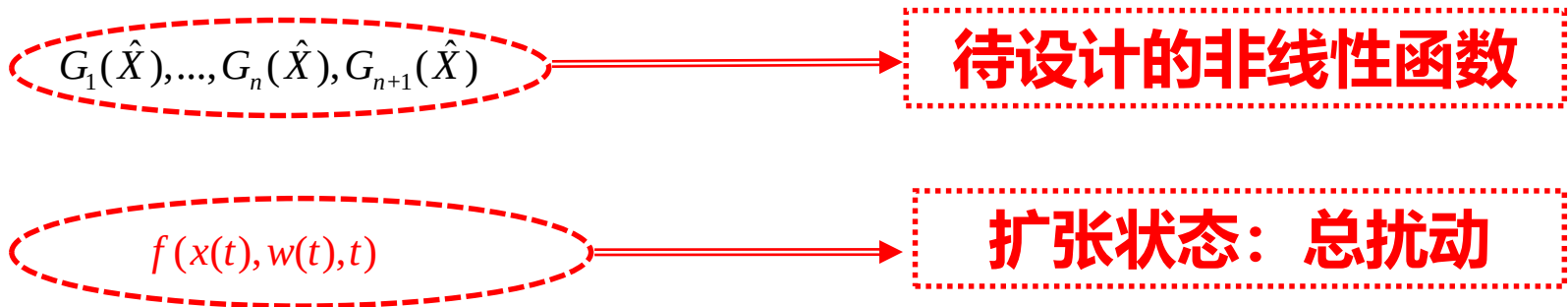
扩张状态观测器：估计纯积分串联型之外的总扰动

韩京清. 一类不确定对象的扩张状态观测器. 控制与决策, 1995, 1: 85-88.

扩张状态观测器
Extended State Observer
(ESO):

$$\begin{cases} \dot{\hat{x}}_1 = \hat{x}_2 - G_1(\hat{x}_1(t) - y(t)) \\ \dots \\ \dot{\hat{x}}_n = \hat{x}_{n+1} - G_n(\hat{x}_1(t) - y(t)) + bu \\ \dot{\hat{x}}_{n+1} = -G_{n+1}(\hat{x}_1(t) - y(t)) \end{cases} \quad (2)$$

$$\hat{x}_1(t) \rightarrow x_1(t), \dots, \hat{x}_n(t) \rightarrow x_n(t), \hat{x}_{n+1}(t) \rightarrow f(x(t), w(t), t) \quad (3)$$



例子：运动控制系统



韩京清 (1937-2008)

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2, w(t), t) + bu \\ y = x_1 \end{cases}$$



$$f(x_1, x_2, w(t), t) = (y^{(2)} - bu)$$

理想的控制器

$$u = \frac{-f(x_1, x_2, w(t), t)}{b} + \varphi(y - r, x_2)$$

可视为系统输出2阶微分信号的未知部分

通过“总扰动”的实时值的在线估计实现补偿

信号的角度

- 将不确定性视为“总扰动”信号而不是非线性模型
- 通过设计扩张状态观测器在线估计“总扰动”实时值

- 韩京清, 一类不确定系统的扩张状态观测器, 控制与决策, 1995, 10(1): 85-88.
- 韩京清, 自抗扰控制及其应用, 控制与决策, 1998, 13(1):19-23.
- J. Han, From PID to Active Disturbance Rejection Control, IEEE TIE, 2009, 56(3): 900-906.

例子：运动控制系统

- **对象**
$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2, w(t), t) + bu \end{cases}$$

$$y = x_1$$

- **标准型** $\ddot{y} = bu$

- **总扰动估计值** $\hat{f}(t) \approx f(x_1, x_2, w(t), t)$

总扰动的估计：扩张状态观测器

$$\ddot{y} = f(x_1, x_2, w(t), t) + bu \xrightarrow{x_3 = f(y, \dot{y}, w, t)} \begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 + bu, \\ \dot{x}_3 = \dot{f} \\ y = x_1 \end{cases}$$

Extended State Observer (ESO)

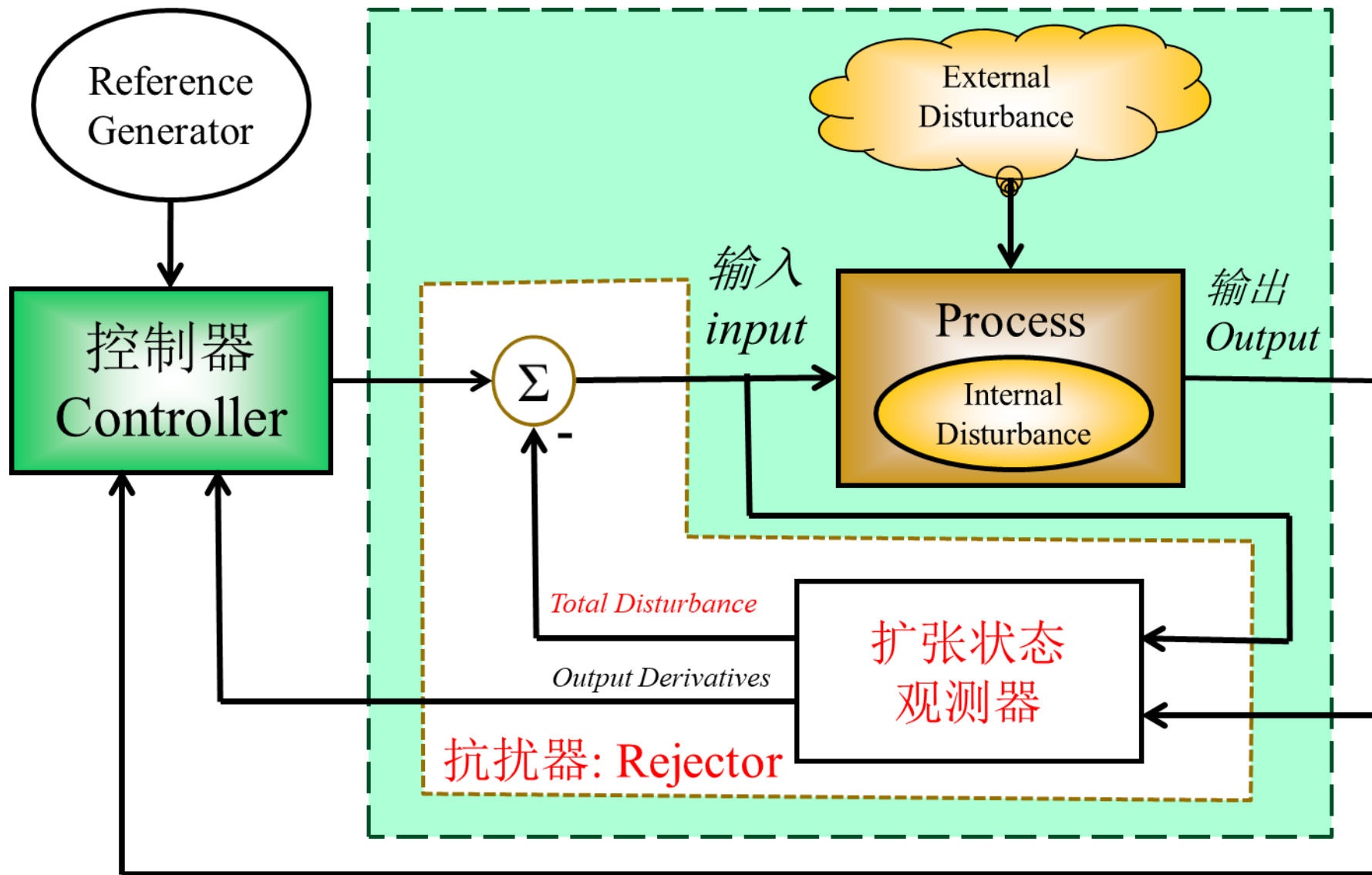
$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - \beta_1 G_1 (\hat{X}_1 - y) \\ \dot{\hat{X}}_2 = \hat{X}_3 - \beta_2 G_2 (\hat{X}_1 - y) + bu \\ \dot{\hat{X}}_3 = -\beta_3 G_3 (\hat{X}_1 - y) \end{cases}$$

$$\hat{X}_1 \approx x_1 \quad \hat{X}_2 \approx x_2 \quad \hat{X}_3 \approx x_3 = f$$

自抗扰设计

$$\ddot{y} = f(x_1, x_2, w(t), t) + bu \quad \xrightarrow{u = u_0 - \frac{\hat{f}}{b}, \hat{f} \approx f} \quad \ddot{y} \approx bu_0$$

自抗扰控制的核心部分：扩张状态观测器



Motion Control Test, 1997, Cleveland State University



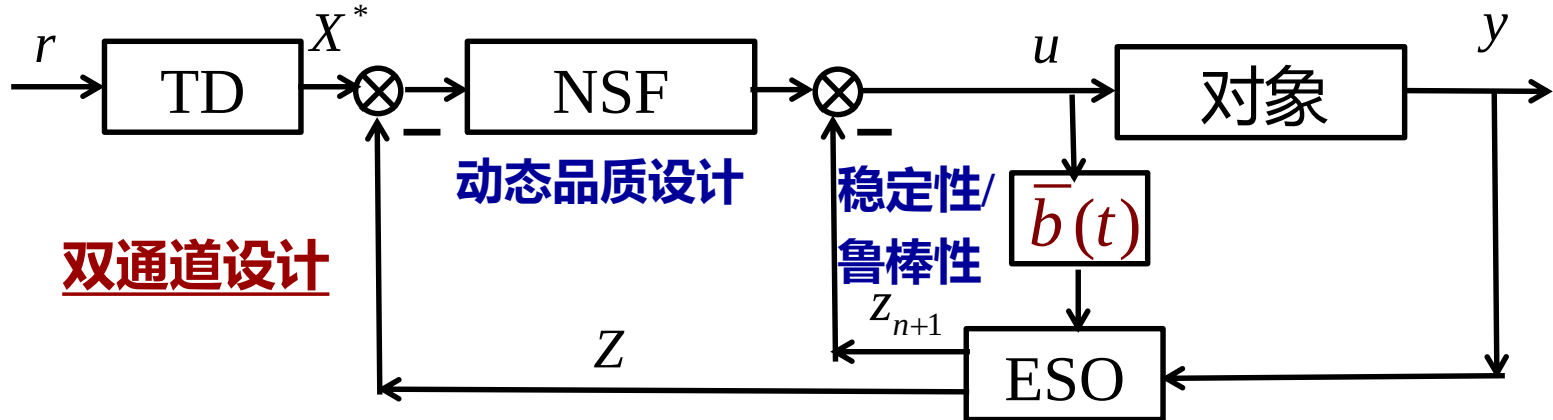
自抗扰控制的思想与基本结构

- 自抗扰控制的思想
 - **自抗扰控制的非线性设计方法**
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-

自抗扰控制 (ADRC)的早期设计框架

- (1) Tracking Differentiator (TD) : 跟踪微分器
- (2) Nonlinear State Feedback (NSF): 非线性状态反馈
- (3) Extended State Observer (ESO): 扩张状态观测器

$$\begin{cases} \dot{x}^{(n)}(t) = f(x, w(t), t) + b(X, t)u(t) \\ y = x(t) \end{cases}, X = [x, \dots, x^{(n-1)}]^T$$



跟踪控制

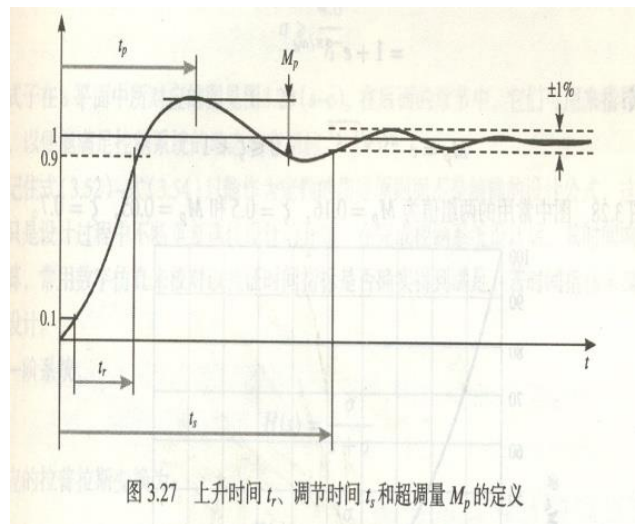
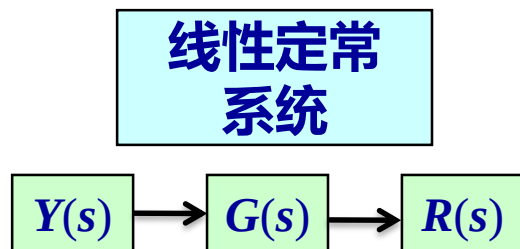
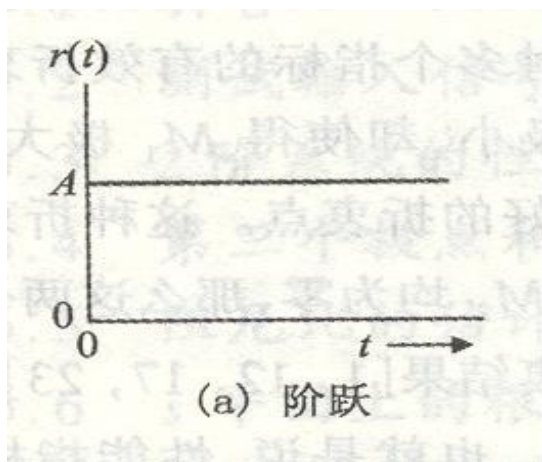
$$u = \frac{-\varphi(\hat{x} - x^*) - \hat{x}_{n+1}}{\bar{b}(t)}, \quad \hat{x} = [\hat{x}_1, \dots, \hat{x}_n]^T$$

$$x^* = [x^*, \dots, x^{(n)*}]^T$$

自动控制原理：系统对阶跃信号的响应

$$Y(s) = \frac{a_2}{s^2 + a_1 s + a_2} R(s), \quad a_1 > 0, a_2 > 0$$

稳定的系统！



控制输入为阶跃信号幅值时，

容易使得系统输入瞬态等达到饱和，控制性能变差！

ADRC第1部分：跟踪微分器(Tracking Differentiator, TD): 安排过渡过程

TD的形式:

$$\begin{cases} \dot{z}_1 = z_2 \\ \dots \\ \dot{z}_{n-1} = z_n \\ \dot{z}_n = r^n h(z_1 - y(t), \frac{z_2}{r}, \dots, \frac{z_n}{r^{n-1}}) \end{cases}$$

被跟踪的信号

调节的参数

- 输入信号的约束条件:

$y(t)$ 在任意有界区间 $t \in [t_0, t_0 + L], 0 < t_0 < t_0 + L < \infty$ 上有界并可积。

- 输出信号的特性:

$$\lim_{r \rightarrow \infty} \int_{t_0}^{t_0+L} |z_1(t) - y(t)| dt = 0$$

当 r 足够大时, $z_1(t)$ 可以充分接近于 $y(t), t \in [t_0, t_0 + L]$

$z_i(t), i \in n$ 视为 $y^{(i-1)}(t), t \in [t_0, t_0 + L]$ 的估计值. 当 $y^{(i-2)}(t)$ 不可导时, $y^{(i-1)}(t)$ 为其广义导

- 韩京清, 王伟. 非线性跟踪 - 微分器. 系统科学与数学, 1994, 14(2): 177-183.
- XUE Wenchao, HUANG Yi, YANG Xiaoxia, What Kinds of System Can Be Used as Tracking-Differentiator, 2009 CCC, 2009.

ADRC第1部分：跟踪微分器(Tracking Differentiator, TD): 安排过渡过程

$$\text{TD: } \begin{cases} \dot{z}_1 = z_2 \\ \dots \\ \dot{z}_{n-1} = z_n \\ \dot{z}_n = r^n h(z_1 - y(t), \frac{z_2}{r}, \dots, \frac{z_n}{r^{n-1}}) \end{cases}$$

$$\text{原始的系统: } \begin{cases} \dot{x}_1 = x_2 \\ \dots \\ \dot{x}_n = h(x_1, x_2, \dots, x_n) \\ y = x_1 \end{cases}$$

被跟踪的信号：假设条件

■ 设输入信号 $y_1(t) \triangleq y(t)$ 在 $[t_0, t_0 + L]$ 上只含有第一类不连续点 $\{t_i\}_{i=1}^{q_L}$ ，且满足：

$$\sup_i |y_1(t_i^-) - y_1(t_i^+)| \leq w_1 < \infty.$$

■ 在 $t \in [t_0, \infty) \setminus \{t_i\}_{i=1}^{q_L}$ 时 $y_2(t) \triangleq \dot{y}_1(t)$ 存在，并且 $y_2(t)$ 在间断点 $\{t_i\}_{i=1}^{q_L}$ 的左右极限

$$y_2(t_i^-), y_2(t_i^+) \text{ 存在以及 } \sup_{t \in [t_0, t_0 + L]} \{|y_2(t^+)|, |y_2(t^-)|\} \leq w_2 < \infty.$$

ADRC第1部分：跟踪微分器(Tracking Differentiator, TD): 安排过渡过程

$$\text{TD:} \begin{cases} \dot{z}_1 = z_2 \\ \dots \\ \dot{z}_{n-1} = z_n \\ \dot{z}_n = r^n h(z_1 - y(t), \frac{z_2}{r}, \dots, \frac{z_n}{r^{n-1}}) \end{cases} \quad \text{原始的系统:} \begin{cases} \dot{x}_1 = x_2 \\ \dots \\ \dot{x}_n = h(x_1, x_2, \dots, x_n) \\ y = x_1 \end{cases}$$

定理： 设原始系统原点在 $(0, \infty) \times \mathcal{B}(0, \rho)$, $\rho > w_1$ 上一致渐近稳定，则当 $[z_1(t_0) - y(t_0), z_2(t_0), \dots, z_n(t_0)]^T \in \mathcal{B}(0, \rho)$ 时，TD满足：

$$\lim_{r \rightarrow \infty} \int_{t_0}^{t_0+L} |z_1(t) - y(t)| dt = 0.$$

最速控制形式（最常用的）的TD

对于二阶线性定常系统：

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = u, |u| \leq r \end{cases}$$

以原点为终点的快速
最优控制综合函数为：

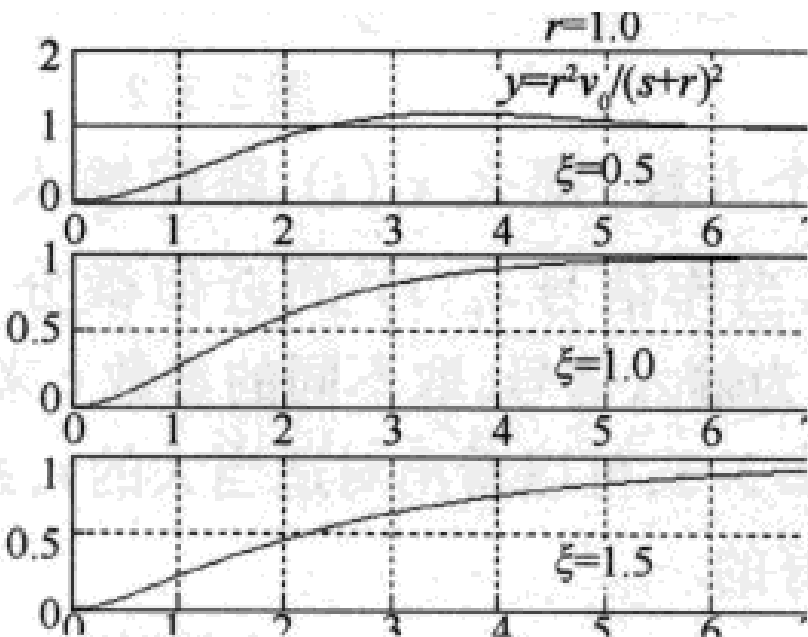
$$u(x_1, x_2) = -r \operatorname{sign}\left(x_1 + \frac{x_2 |x_2|}{2r}\right)$$

快速最优控制对应的TD为：

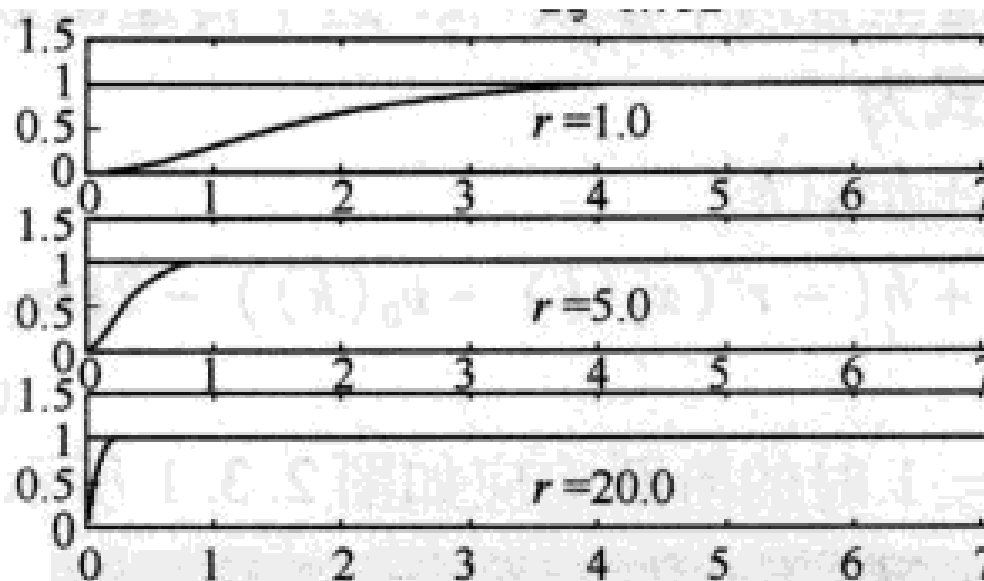
$$\begin{cases} \dot{z}_1 = z_2 \\ \dot{z}_2 = -r \operatorname{sign}\left(z_1 - y(t) + \frac{z_2 |z_2|}{2r}\right) \end{cases}$$

这个系统的解的分量 $z_1(t)$ 在加速度限制 $|\ddot{z}_1| \leq r$ 之下，将最快地跟踪输入信号 $y(t)$ ，而且 r 越大，跟踪得越快。显然，当 $z_1(t)$ 充分地接近 $y(t)$ 时，可以把解的另一分量 $z_2(t) = \dot{z}_1(t)$ 当作输入信号 $y(t)$ 的近似微分。

二阶惯性环节安排 过渡过程



TD 安排过渡过程



韩京清，自抗扰控制技术，国防工业出版社，2008.

ADRC第2部分：非光滑状态反馈

先考察一个简单例子：设有一阶受控对象

$$\dot{x} = f(x, t) + u$$

对它施以状态的线性反馈

$$u = -kx, k > 0.$$

得闭环系统

$$\dot{x} = -kx + f(x, t)$$

今取上述状态反馈,使其状态变量 x 在扰动 $f(x, t)$ 的作用下,尽可能地收敛到零,这里的 k 称作**反馈增益**。

进一步假定：存在两常数 m, w_0 , 满足 $|f(x, t)| < \left| \frac{x}{m} \right| + w_0, \quad \forall x \in R, \forall t > 0$

线性反馈

$$\dot{x} = -kx + f(x, t) \quad (2.1)$$

$$|f(x, t)| < \left| \frac{x}{m} \right| + w_0, \forall x, t > 0$$

对于系统(2.1), 有

(1) 若 $w_0 = 0$, 而取反馈增益 $k > \frac{1}{m}$, 那么对所有 x 都有 $\frac{dx^2}{dt} < 0$,
系统(2.1)渐近稳定, 所有解都趋近于0.

线性反馈

$$\dot{x} = -kx + f(x, t) \quad (2.1)$$

$$|f(x, t)| < \left| \frac{x}{m} \right| + w_0, \forall x, t > 0$$

对于系统(2.1), 有

(1) 若 $w_0 = 0$, 而取反馈增益 $k > \frac{1}{m}$, 那么对所有 x 都有 $\frac{dx^2}{dt} < 0$, 系统(2.1)渐近稳定, 所有解都趋近于0.

(2) 若 $w_0 \neq 0$, 而取反馈增益 $k > \frac{1}{m}$, 那么当 $|x| > \frac{m}{m-1/k} \frac{w_0}{k}$ 时, 就有 $\frac{dx^2}{dt} < 0$, 因此, 闭环系统(2.1)的解满足

$$\lim_{t \rightarrow \infty} x(t) \in \left[-\frac{m}{m-1/k} \frac{w_0}{k}, \frac{m}{m-1/k} \frac{w_0}{k} \right]$$

非光滑反馈

非光滑控制反馈: $u = -k |x|^\alpha \operatorname{sign}(x), 0 < \alpha < 1$

闭环系统: $\dot{x} = -k |x|^\alpha \operatorname{sign}(x) + f(x, t) \quad (2.2)$

进一步假定: 存在两常数 m, w_0 , 满足 $|f(x, t)| < \left| \frac{x}{m} \right|^\alpha + w_0, \forall x, t > 0$

非光滑反馈

非光滑控制反馈: $u = -k |x|^\alpha \operatorname{sign}(x), 0 < \alpha < 1$

闭环系统: $\dot{x} = -k |x|^\alpha \operatorname{sign}(x) + f(x, t) \quad (2.2)$

进一步假定: 存在两常数 m, w_0 , 满足 $|f(x, t)| < \left| \frac{x}{m} \right|^\alpha + w_0, \forall x, t > 0$

对于系统(2.2), 可以得出

$$\begin{aligned} \frac{d|x|^2}{dt} &= -2k |x|^{\alpha+1} \operatorname{sign}(x) + 2xf(x, t) \\ &= -2k |x| \left(|x|^\alpha - \operatorname{sign}(x) \frac{f(x, t)}{k} \right) \\ &< -2k |x| \left(|x|^\alpha - \frac{1}{k} \left(\left| \frac{x}{m} \right|^\alpha + w_0 \right) \right) \end{aligned}$$

非光滑反馈

非光滑控制反馈: $u = -k |x|^\alpha \operatorname{sign}(x), 0 < \alpha < 1$

闭环系统: $\dot{x} = -k |x|^\alpha \operatorname{sign}(x) + f(x, t) \quad (2.2)$

$$|f(x, t)| < \left| \frac{x}{m} \right|^\alpha + w_0, \forall x, t > 0$$

选取反馈增益 $k > 1/m^\alpha$. 进一步因为

$$|x|^\alpha > \frac{m^\alpha}{m^\alpha - 1/k} \frac{w_0}{k} \Rightarrow \left(|x|^\alpha - \frac{1}{k} \left(\left| \frac{x}{m} \right|^\alpha + w_0 \right) \right) > 0 \Rightarrow \frac{d|x|^2}{dt} < 0$$

所以(2.2)满足:

$$\lim_{t \rightarrow \infty} x(t) \in \left[- \left(\frac{m^\alpha}{m^\alpha - 1/k} \right)^{\frac{1}{\alpha}} \left(\frac{w_0}{k} \right)^{\frac{1}{\alpha}}, \left(\frac{m^\alpha}{m^\alpha - 1/k} \right)^{\frac{1}{\alpha}} \left(\frac{w_0}{k} \right)^{\frac{1}{\alpha}} \right]$$

线性反馈与非光滑反馈

在通常情形都有 $k > |w_0|$ ，因此只要

$$k > |w_0|, 0 \leq \alpha < 1, \left(\frac{1}{\alpha} > 1\right),$$

就有

$$\left(\frac{w_0}{k}\right)^{\frac{1}{\alpha}} \ll \frac{w_0}{k}$$

稳态误差**远小于**线性反馈留下的稳态误差。

$$u = -kx, k > 0.$$

从以上的讨论可以看出，在反馈系统中有意识的引入经合适的非线性结构，特别是非光滑结构，将**显著性改善闭环系统的动态特性**。因此开发利用非线性（特别是非光滑）特性是反馈系统设计的很有意义的新课题。

ADRC的核心部分：非线性ESO设计

$$\begin{cases} \dot{\hat{x}}_1 = \hat{x}_2 - G_1(\hat{x}_1(t) - y(t)) \\ \dots \\ \dot{\hat{x}}_n = \hat{x}_{n+1} - G_n(\hat{x}_1(t) - y(t)) + bu \\ \dot{\hat{x}}_{n+1} = -G_{n+1}(\hat{x}_1(t) - y(t)) \end{cases} \quad (2)$$

此系统各状态分别跟踪被扩张的状态变量，即有

$$\hat{x}_1(t) \rightarrow y(t) = x_1(t), \dots, \hat{x}_n(t) \rightarrow y^{(n-1)}(t) = x_n(t), \hat{x}_{n+1}(t) \rightarrow y^{(n)}(t) - bu = x_{n+1}(t) \quad (3)$$

需要通过适当选取的非线性函数 $G_1(\hat{x}), \dots, G_n(\hat{x}), G_{n+1}(\hat{x})$ ，实现上述跟踪目的。

二阶系统的常用非线性ESO

对非线性系统

$$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = f(x_1, x_2, w(t), t) + bu, \\ y = x_1, \end{cases} \quad (2.3)$$

我们把作用于开环系统的加速度 $f(x_1, x_2, w(t), t)$ 的实时作用量扩充成新的状态变量 x_3 ,

$$x_3(t) \triangleq f(x_1, x_2, w(t), t)$$

并记 $w_0(t) \triangleq \dot{x}_3(t)$, 那么系统(2.3)可扩张成新的线性控制系统:

$$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = x_3 + bu, \\ \dot{x}_3 = w_0(t), \\ y = x_1, \end{cases} \quad (2.4)$$

二阶系统的常用非线性ESO

对上述被扩张的系统(2.4)建立状态观测器:

$$\begin{cases} e_1 = \hat{X}_1 - y, \\ \dot{\hat{X}}_1 = \hat{X}_2 - \beta_{01}e_1, \\ \dot{\hat{X}}_2 = \hat{X}_3 - \beta_{02}|e_1|^{1/2} \text{sign}(e_1) + bu, \\ \dot{\hat{X}}_3 = -\beta_{03}|e_1|^{1/4} \text{sign}(e_1), \end{cases} \quad (2.5)$$

$$\hat{X}_1(t) \rightarrow x_1(t),$$

$$\hat{X}_2(t) \rightarrow x_2(t),$$

$$\hat{X}_3(t) \rightarrow x_3(t) = f(x_1, x_2, w(t), t)$$

二阶系统的常用非线性ESO

对上述扩张状态观测器(2.5)来说, 适用于如下三种类型系统

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(t) + bu; \\ y = x_1 \end{cases} \quad \begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2) + bu; \\ y = x_1 \end{cases} \quad \begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2, w(t)) + bu. \\ y = x_1 \end{cases}$$

估计误差方程:
$$\begin{cases} \dot{e}_1 = e_2 - \beta_{01}e_1, & e_1 = \hat{X}_1 - x_1 \\ \dot{e}_2 = e_3 - \beta_{02}|e_1|^{1/2} \text{sign}(e_1), & e_2 = \hat{X}_2 - x_2 \\ \dot{e}_3 = w_0 - \beta_{03}|e_1|^{1/4} \text{sign}(e_1), & e_3 = \hat{X}_3 - x_3 \end{cases}$$

假设 $\lim_{t \rightarrow \infty} w_0(t) = w^*$, 则系统进入稳态时, 即方程右端全收敛于零时, 得到稳态误差为:

$$\lim_{t \rightarrow \infty} |e_1(t)| = \left(\frac{w^*}{\beta_{03}} \right)^4, \quad \lim_{t \rightarrow \infty} |e_2(t)| = \beta_{01} \left(\frac{w^*}{\beta_{03}} \right)^4, \quad \lim_{t \rightarrow \infty} |e_3(t)| = \beta_{02} \left(\frac{w^*}{\beta_{03}} \right)^2$$

二阶系统的常用非线性ESO

在进行数值仿真时，为了避免高频颤振现象的出现，我们把函数 $|e|^\alpha \text{sign}(e)$ 改造成原点附近具有线性段的连续的幂次函数

$$\text{fal}(e, \alpha, \delta) = \begin{cases} \frac{e}{\delta^{n-1}}, & |e| \leq \delta \\ |e|^\alpha \text{sign}(e), & |e| > \delta \end{cases}$$

式中： δ 为线性段的区间长度，如图4.3.3所示。

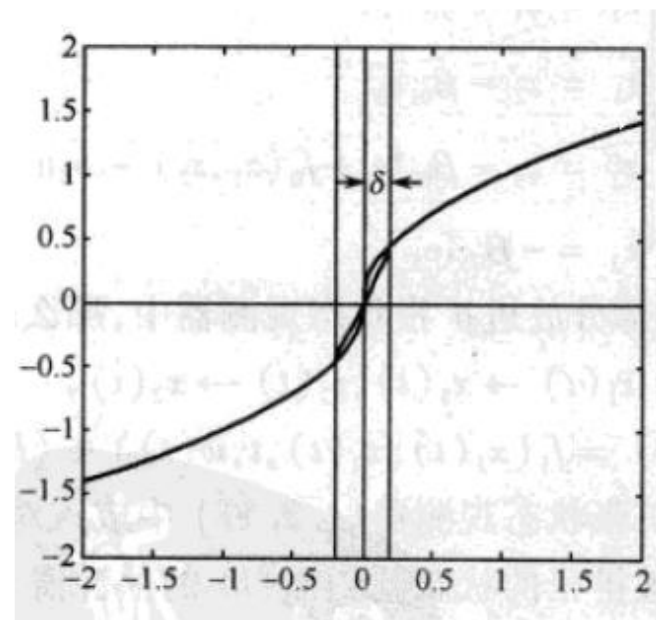


图4.3.3连续的幂次函数

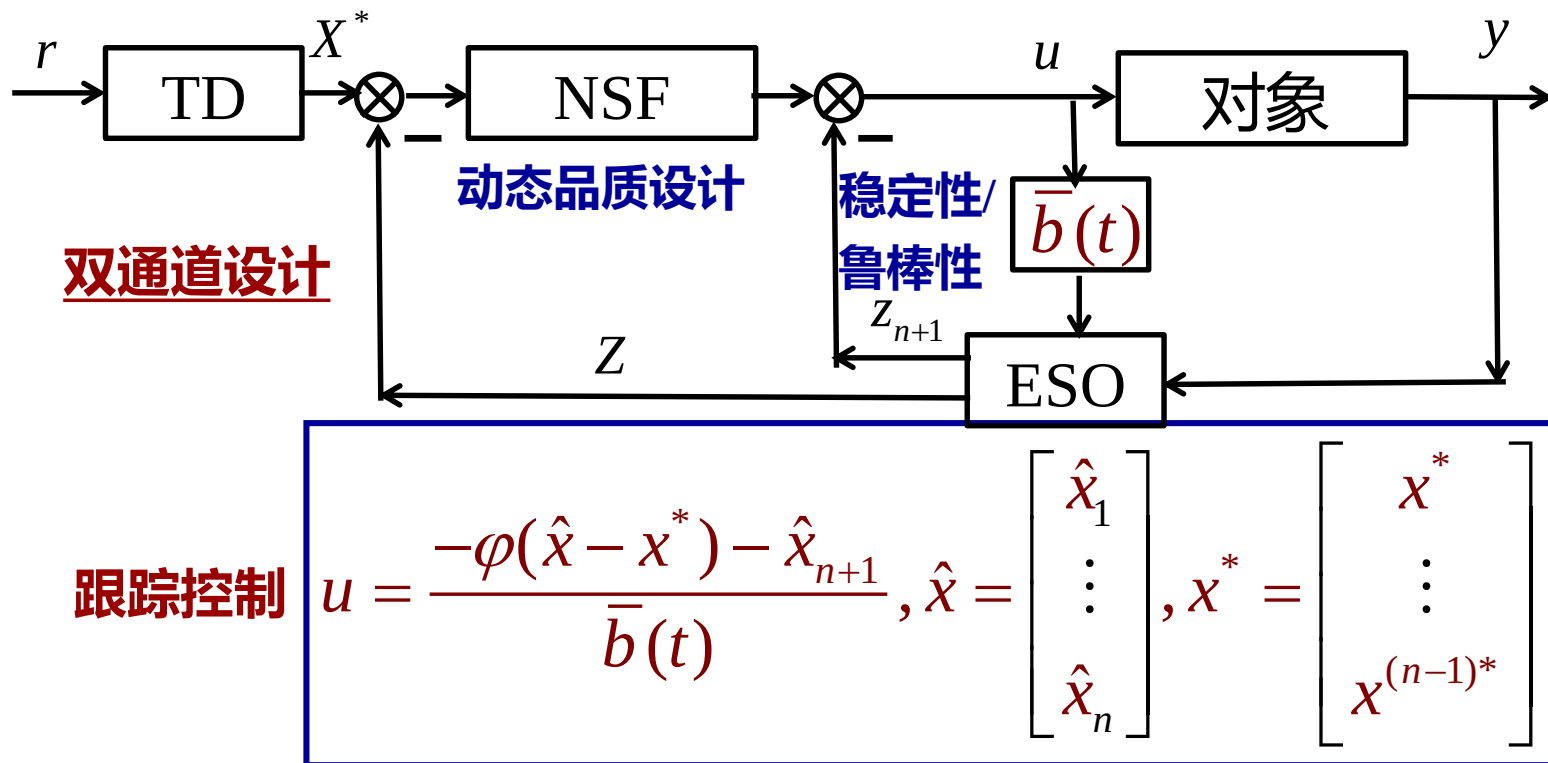
**常用的
非线性ESO设计**

$$\begin{cases} \dot{\hat{x}}_1 = \hat{x}_1 + h(\hat{x}_2 - \beta_{01}e) \\ \dot{\hat{x}}_2 = \hat{x}_2 + h(\hat{x}_3 - \beta_{02}f_e) \\ \dot{\hat{x}}_3 = \hat{x}_3 + h(-\beta_{03}f_{e1}) \\ e = \hat{x}_1 - y, \quad f_e = \text{fal}(e, \frac{1}{2}, \delta), \quad f_{e1} = \text{fal}(e, \frac{1}{4}, \delta) \end{cases}$$

自抗扰控制 (ADRC)的早期设计框架

非线性设计较多

$$\begin{cases} x^{(n)}(t) = f(x, w(t), t) + b(X, t)u(t) \\ y = x(t) \end{cases}, X = \begin{bmatrix} x \\ \vdots \\ x^{(n-1)} \end{bmatrix} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$



自抗扰控制的思想与基本结构

- 自抗扰控制的思想
 - 自抗扰控制的非线性设计方法
 - **自抗扰控制的线性设计方法**
 - 自抗扰控制与PID控制的关系
 - 自抗扰控制的一些应用实例
-

线性ADRC

$$\begin{cases} \dot{X}_1 = X_2 \\ \dots \\ \dot{X}_{n-1} = X_n \\ \dot{X}_n = \mathbf{X}_{n+1}(\bullet) + \bar{B}(t)U(t) \end{cases}$$

A parameterized linear ADRC (LADRC) was proposed in Gao Z. (2003, CDC)

使得控制器参数具有物理意义，更加易于调节

**非线性
ADRC**

ESO

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - G_1(\hat{X}_1 - Y) \\ \dots \\ \dot{\hat{X}}_n = \hat{X}_{n+1} - G_n(\hat{X}_1 - Y) + \bar{B}(t)U(t) \\ \dot{\hat{X}}_{n+1} = -G_{n+1}(\hat{X}_1 - Y) \end{cases}$$

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} + U_0(Y, \hat{X}_2, \dots, \hat{X}_n)$$

LADRC

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - \beta_1(\hat{X}_1 - Y) \\ \dots \\ \dot{\hat{X}}_n = \hat{X}_{n+1} - \beta_n(\hat{X}_1 - Y) + \bar{B}(t)U(t) \\ \dot{\hat{X}}_{n+1} = -\beta_{n+1}(\hat{X}_1 - Y) \end{cases}$$

增益：极点配置

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} - \underbrace{\bar{B}^{-1}(t)\sum_{i=1}^n k_i \hat{X}_i}_{\text{闭环动态设计：极点配置}}$$

闭环动态设计：极点配置

预期系统的响应

带宽参数化的线性ADRC

$$\begin{cases} \dot{X}_1 = X_2 \\ \dots \\ \dot{X}_{n-1} = X_n \\ \dot{X}_n = \mathbf{X}_{n+1}(\bullet) + \bar{B}(t)U(t) \end{cases}$$

A parameterized linear ADRC (LADRC) was proposed in Gao Z. (2003, CDC)

使得控制器参数具有物理意义，更加易于调节

非线性
ADRC

ESO

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - G_1(\hat{X}_1 - Y) \\ \dots \\ \dot{\hat{X}}_n = \hat{X}_{n+1} - G_n(\hat{X}_1 - Y) + \bar{B}(t)U(t) \\ \dot{\hat{X}}_{n+1} = -G_{n+1}(\hat{X}_1 - Y) \end{cases}$$

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} + U_0(Y, \hat{X}_2, \dots, \hat{X}_n)$$

LADRC

$$\begin{cases} \dot{\hat{X}}_1 = \hat{X}_2 - \beta_1(\hat{X}_1 - Y) \\ \dots \\ \dot{\hat{X}}_n = \hat{X}_{n+1} - \beta_n(\hat{X}_1 - Y) + \bar{B}(t)U(t) \\ \dot{\hat{X}}_{n+1} = -\beta_{n+1}(\hat{X}_1 - Y) \end{cases}$$

$$s^{n+1} + \sum_{i=1}^{n+1} \beta_i s^{n-i+1} = (s + \omega_e)^{n+1} \quad \omega_e > 0$$

LESO带宽

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} - \bar{B}^{-1}(t) \sum_{i=1}^n k_i \hat{X}_i$$

闭环动态设计：极点配置

$$s^n + \sum_{i=1}^n k_i s^{i-1} = (s + \omega_c)^n, \quad \omega_c > 0$$

预期动态的带宽

ESO (1995)

$$\begin{cases} \dot{X}_1 = X_2 \\ \dots \\ \dot{X}_{n-1} = X_n \\ \dot{X}_n = \mathbf{X}_{n+1} + \bar{B}(t)U(t) \end{cases}$$

标准型 + 总扰动

$$\begin{cases} \hat{X}_1 \rightarrow X_1 = Y \\ \dots \\ \hat{X}_n \rightarrow X_n = Y^{(n-1)} \\ \hat{X}_{n+1} \rightarrow X_{n+1} \end{cases}$$

ADRC (Han,1998)

$$U(t) = -\bar{B}^{-1}(t)\hat{X}_{n+1} + U_0(Y, \hat{X}_2, \dots, \hat{X}_n)$$

抵消总扰动的效应

鲁棒性

对标称系统的控制

预期的性能

$$\begin{cases} \dot{X}_1 = X_2 \\ \dots \\ \dot{X}_{n-1} = X_n \\ \dot{X}_n = \bar{B}(t)U_0(\bullet) \end{cases}$$

双自由度设计

自抗扰控制的思想与基本结构

- 自抗扰控制的思想
 - 自抗扰控制的非线性设计方法
 - 自抗扰控制的线性设计方法
 - **自抗扰控制与PID控制的关系**
 - 自抗扰控制的一些应用实例
-

PID (Proportional-Integral-Derivative) 控制

典型运动控制系统

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2, w(t), t) + bu \\ y = x_1 \end{cases}$$

不确定动态: $f(\cdot)$

外部扰动: $w(t)$

位置: x_1 速度: x_2

量测输出: y

控制输入: u

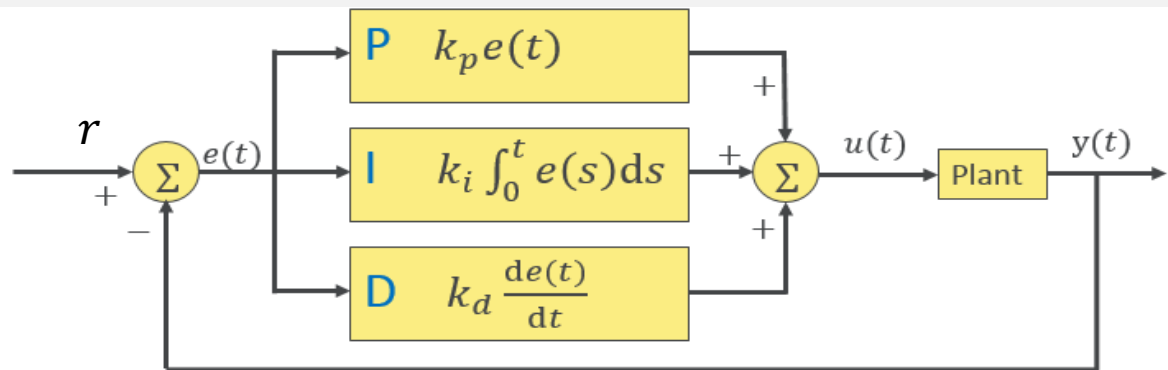
控制目标: $\lim_{t \rightarrow \infty} y = r$ (设定点)

PID 控制器

工业中90%以上的控制器

$$u = k_p e + k_i \int e dt + k_d \dot{e}$$

$$e \triangleq r - y \quad (\text{tracking error})$$



[1] K. J. Astrom and T. Hagglund, Advanced PID control, North Carolina, 2006.

[2] W. L. Bialkowski, The control handbook, New York, 1996.

[3] C. Zhao and L. Guo, PID controller design for second order *nonlinear uncertain systems*, SCIS, 2017.

[4] C. Zhao and L. Guo, Towards a Theoretical Foundation of *PID Control for Uncertain Nonlinear Systems*, Automatica, 2022.

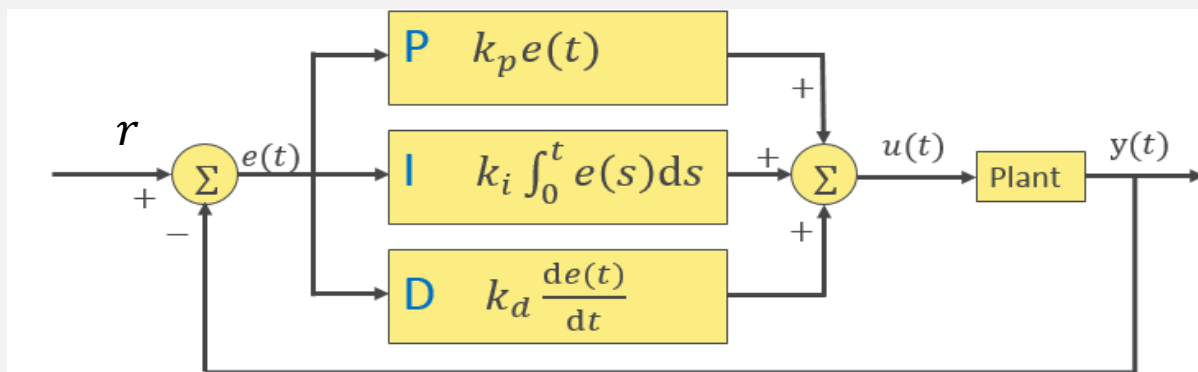
PID 控制的优点

PID: 比例（现在），积分（过去），微分（将来）

PID控制的形式

$$u = k_p e + k_I \int e dt + k_D \dot{e}$$

$$e \triangleq r - y \quad (\text{tracking error})$$



PID控制算法：统治世界的十大算法之一

- ◆ 百年历史，超过95%的过程控制回路为PID，**没有PID就没有现代文明**
- ◆ 对广泛存在的大范围非线性不确定系统，**缺乏稳定性及参数设计的理论基础**

基于误差来消除误差的控制方法

□ 数据驱动机制: 不依赖于系统模型; □ 参数意义明确: 比例、积分和微分增益

二阶非线性不确定系统的PID控制理论

基本的非线性受控力学系统

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2) + u(t) \end{cases}$$

跟踪误差: $e(t) = y^* - x_1(t)$

$$u(t) = k_p e(t) + k_i \int_0^t e(\tau) d\tau + k_d \dot{e}(t)$$

控制目的: $\lim_{t \rightarrow \infty} e(t) = 0$

非线性不确定性函数 $f(\cdot)$ 集合的度量:

$$\mathcal{F}_{L_1, L_2} = \left\{ f(\cdot) \left| \frac{\partial f}{\partial x_1} \leq L_1, \left| \frac{\partial f}{\partial x_2} \right| \leq L_2, \forall x_1, x_2 \in \mathbb{R} \right. \right\}$$

$\mathcal{F}_{L_1, L_2}$

$f(\cdot)$



二阶非线性不确定系统的PID控制理论（续）

定理1（单输入单输出情况）： 当PID参数在下述集合中选取时，

$$\Omega_{pid} = \left\{ \begin{bmatrix} k_p \\ k_i \\ k_d \end{bmatrix} \mid k_p > L_1, k_i > 0, (k_p - L_1)(k_d - L_2) > k_i + L_2 \sqrt{k_i(k_d + L_2)} \right\}$$

对任意 $f(\cdot) \in \mathcal{F}_{L_1 L_2}$, 任意 $(x_1(0), x_2(0)) \in \mathbb{R}^2$, 任意 $y^* \in \mathbb{R}$, 有

$$\lim_{t \rightarrow \infty} e(t) = 0, \lim_{t \rightarrow \infty} x_2(t) = 0$$

**针对二阶非线性不确定系统，
定量地给出了的PID控制器参数选取范围**

Cheng Zhao, and Lei Guo, *SCIENCE CHINA Information Sciences*, 2017.

二阶非线性不确定系统的PID控制理论（续）

定理2（多输入多输出情况）： 当PID参数矩阵 (k_p, k_i, k_d) 属于如下集合：

$$\Omega_{pid} = \{\bar{k}_p, k_i, \bar{k}_d \in \mathbb{R}^{3+} \mid \bar{k}_p \bar{k}_d - k_i > \sqrt{(\bar{k}_p L_1 + k_i L_2)(L_1 + \bar{k}_d L_2)}\}$$

$\bar{k}_p = k_p - L_1$, $\bar{k}_d = k_d - L_2$, 那么对任意的 $f \in \mathcal{F}_{L_1, L_2}$, 任意的系统初值和任意的输出目标向量 y^* , 闭环控制系统都是稳定的, 并且跟踪误差以指数级收敛速度满足 $\lim_{t \rightarrow \infty} e(t) = 0$.

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2) + u(t) \end{cases} \quad u(t) = k_p e(t) + k_i \int_0^t e(\tau) d\tau + k_d \dot{e}(t)$$

非线性不确定函数向量 $f(x_1, x_2)$ 的度量：

$$\mathcal{F}_{L_1, L_2} = \{f \in C^1(\mathbb{R}^{2n} \times \mathbb{R}^+, \mathbb{R}^n) \mid \left\| \frac{\partial f}{\partial x_1} \right\| \leq L_1, \left\| \frac{\partial f}{\partial x_2} \right\| \leq L_2\}, L_1, L_2 > 0.$$

Jinke Zhang, and Lei Guo, *Theory and Design of PID Controller for Nonlinear Uncertain Systems*, *IEEE Control Systems Letters*, vol. 3, no. 3, pp. 643-648, 2019.



进一步的问题：

从集合 Ω_{pid} 中选取特定的PID参数，实现
在线估计和补偿不确定性项 $f(x_1, x_2)$ ，提高动态性能

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2) + u(t) \end{cases} \quad u(t) = k_p e(t) + k_i \int_0^t e(\tau) d\tau + k_d \dot{e}(t)$$

利用自抗扰控制方法的PID调参律

体现实时估计和补偿复杂非线性不确定动态及扰动产生的效应，
实现预期的闭环系统动态性能。

自抗扰控制型的PID调参方法

基本的非线性受控力学系统

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2) + u(t) \end{cases}$$

跟踪误差: $e(t) = y^* - x_1(t)$

$$u(t) = k_p e(t) + k_i \int_0^t e(\tau) d\tau + k_d \dot{e}(t)$$

控制目的: $\lim_{t \rightarrow \infty} e(t) = 0$

建立典型自抗扰控制设计与PID控制的关系

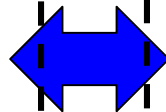
自抗扰控制设计

降阶
扩张状态观
测器

$$\begin{cases} \dot{\xi} = -\omega_o \xi - \omega_o^2 y_2 - \omega_o u, \\ \hat{f}_t = \xi + \omega_o y_2. \end{cases}$$

实时估计和补偿 $f(x_1, x_2)$

$$u = -k_{ap}(y_1 - r) - k_{ad}(y_2 - \dot{r}) - \hat{f}_t + \ddot{r}$$



等价的PID形式

$$u = -k_p e - k_d \dot{e} - k_i \int_0^t e(\tau) d\tau + \ddot{r}$$

$$k_p = k_{ap} + \omega_o k_{ad},$$

$$k_d = k_{ad} + \omega_o,$$

$$k_i = \omega_o k_{ap}.$$

自抗扰控制型的PID调参方法

自抗扰型PID调参方法：

$$k_p = k_{ap} + \omega_o k_{ad},$$

$$k_d = k_{ad} + \omega_o,$$

$$k_i = \omega_o k_{ap}.$$

PID控制器的
3个参数

典型自抗扰控制器的
3个参数

➤ 给出了PID控制器对不确定性项 $f(x_1, x_2)$ 的在线估计值：

P-I-D三项的线性组合： $\hat{f}_t = -\omega_0 \left\{ k_{ad} e(t) + k_{ap} \int_0^t e(s) ds + \dot{e}(t) \right\}$

自抗扰控制型的PID调参方法

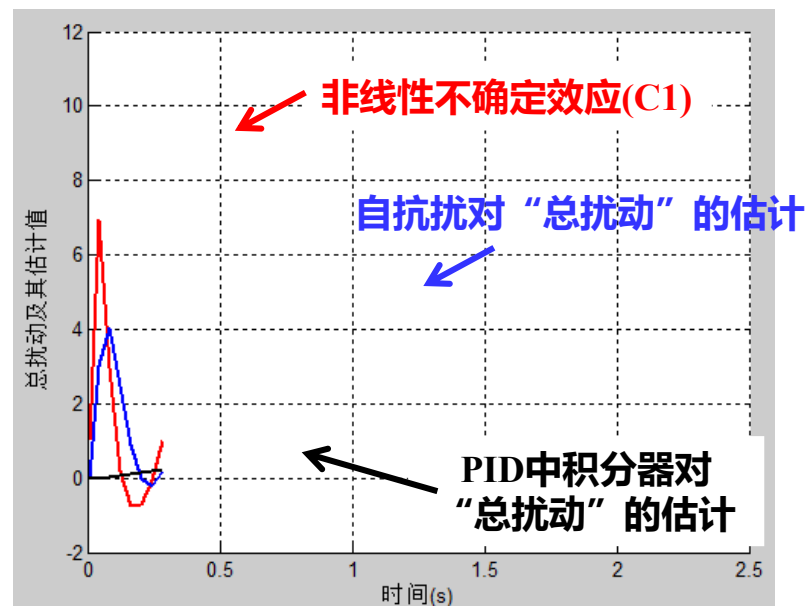
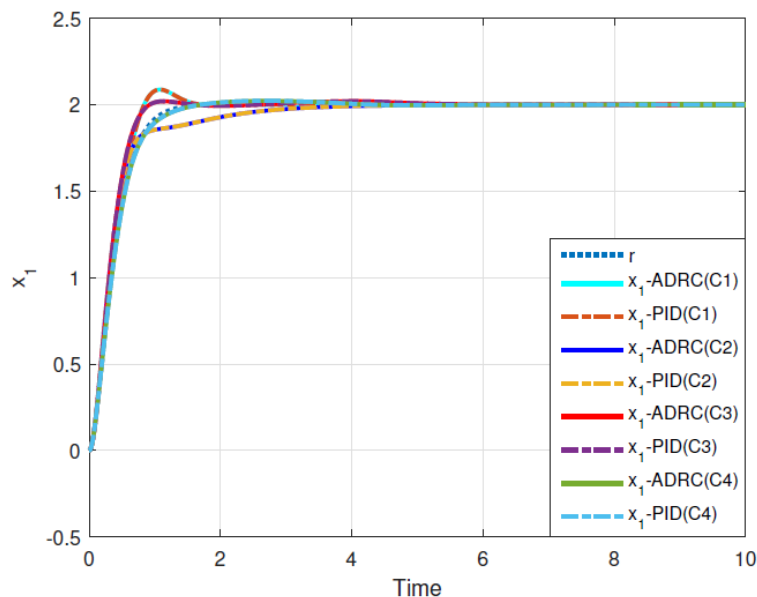
自抗扰的扰动估计值一定快于PID中积分器的扰动估计值

$$\hat{F}_I(s) = \frac{k_{ap}}{s^2 + k_{ad}s + k_{ap}} \hat{F}(s)$$

PID积分器的估计

自抗扰中扩张状态观测器的估计
(P-I-D三项的线性组合)

仿真例子: $\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(x_1, x_2) + u(t) \end{cases}$ $C1: f(t) = 2x_1 + 6\sin x_2 + 1, \quad C2: f(t) = 5\cos x_1 + 2x_2 - 2,$
 $C3: f(t) = 3x_1 + 2x_2 - w_1(t), \quad C4: f(t) = 6\cos x_1 \sin x_2 - w_2(t),$



自抗扰控制型的PID调参方法

自抗扰控制器参数设计的定量结果

定理4. 只要 $\omega_o > \omega_o^*$, 对任意 $f(\cdot) \in \mathcal{F}_{L_1 L_2}$, 任意 $(x_1(0), x_2(0)) \in \mathbb{R}^2$, 任意 $y^* \in \mathbb{R}$, 存在常数 η_1, η_2 , 使得闭环系统有如下性质:

(1) $\lim_{t \rightarrow \infty} x_1(t) = y^{**}, \lim_{t \rightarrow \infty} x_2(t) = 0.$

(2) $|\ddot{e}(t) + k_{ad}\dot{e}(t) + k_{ap}e(t)| \leq \eta_1 e^{-\omega_o t} + \frac{\eta_2}{\omega_o}, \quad t \geq 0.$

$$\omega_o^* = \max \left\{ 0, \frac{L_1 - k_{ap}}{k_{ad}}, L_2 - k_{ad}, \bar{\omega}_o \right\}, \bar{\omega}_o = \begin{cases} 0, & \Omega = \emptyset \text{ or } \max\{\Omega\} \leq 0 \\ \max\{\Omega\}, & \max\{\Omega\} > 0 \end{cases},$$

$$\Omega = \{\omega \in R \mid n_0 \omega^4 + n_1 \omega^3 + n_2 \omega^2 + n_3 \omega + n_4 = 0\}, n_0 = k_{ad}^2$$

$$n_1 = 2k_{ad}[k_{ad}(k_{ad} - L_2) - L_1],$$

$$n_2 = 2k_{ad}(k_{ap} - L_1)(k_{ad} - L_2) + [k_{ad}(k_{ad} - L_2) - L_1]^2 - L_2^2 k_{ap},$$

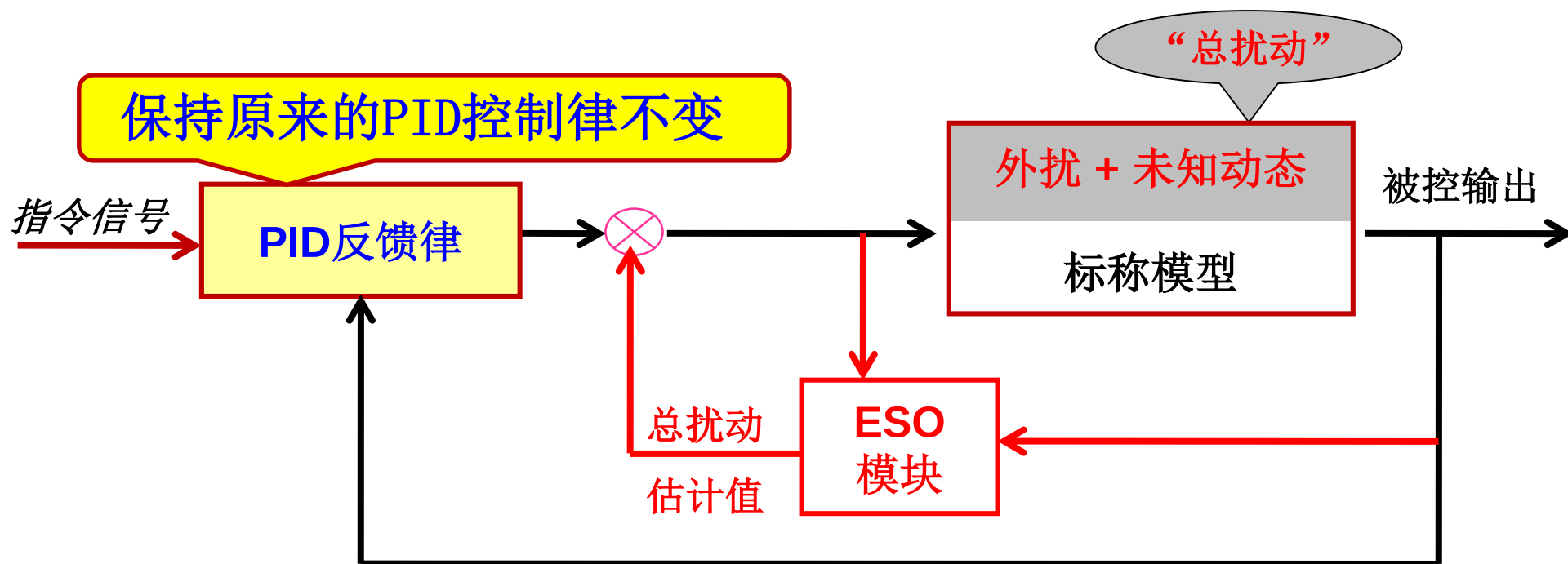
$$n_3 = 2[k_{ad}(k_{ad} - L_2) - L_1](k_{ap} - L_1)(k_{ad} - L_2) - L_2^2 k_{ap}(k_{ad} - L_2),$$

$$n_4 = (k_{ap} - L_1)^2 (k_{ad} - L_2)^2.$$

Zhong S, Huang Y, Guo L. A Parameter Formula Connecting PID and ADRC. Sci China Inf Sci, 2020.



自抗扰控制与PID控制的结合



**对于实际中已使用的PID回路及其参数，
设计“自抗扰控制模块”提高系统对付不确定性的能力**



自抗扰控制与PID控制的结合

对于实际中已使用的PI控制回路及其参数，
设计“自抗扰控制模块”提高系统对付不确定性的能力

一类常见MIMO系统： $\dot{X}(t) = f(X) + D(t) + BU(t), \quad X(t_0) = X_0, X(t) \in R^n$

已使用的PI控制回路： $U(t) = -\bar{K}_p E(t) - \bar{K}_i Z(t), \quad Z(t) \triangleq \int_{\tau=0}^t E(\tau) d\tau, \quad t \geq 0,$

3种不确定性范围

的度量：

$$\left\{ \begin{array}{l} B \in \mathcal{B}_B^q \triangleq \{B \in R^{m \times m} : \|\tilde{B} - I\| \leq q < 1\}, \tilde{B} \triangleq B\bar{B}^{-1}, \\ D(t) \in \mathcal{D}_h^d \triangleq \left\{ D(t) \in \mathcal{C}^1(R \rightarrow R^m) : \right. \\ \quad \left. \|D(t_0)\| \leq d, \left\| \sup_{t \in [t_0, \infty)} \dot{D}(t) \right\| \leq h \right\}, \\ F(\cdot) \in \mathcal{F}_{g(\rho)} \triangleq \left\{ F(\cdot) \in \mathcal{C}^1(R^m \rightarrow R^m) : F(0) = 0, \right. \\ \quad \left. \sup_{\|X\| \leq \rho} \left\| \frac{\partial F}{\partial X} \right\| \leq g(\rho), \quad \rho \geq 0 \right\} \end{array} \right.$$

W. Xue, etc., On Integrating Uncertainty Estimator into PI Control for A Class of Nonlinear Uncertain Systems, IEEE Transactions on Automatic Control, 2020.

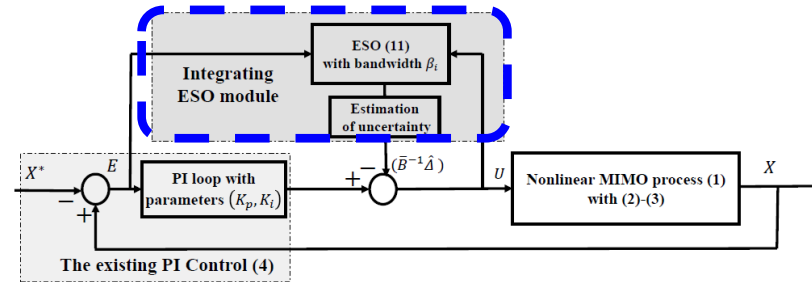


自抗扰控制与PID控制的结合

扩张状态观测器设计:

$$\begin{cases} \dot{\xi}(t) = -\beta\xi(t) - \beta^2 E(t) - \beta\bar{B}U(t), \\ \hat{\Delta}(t) = \beta E(t) + \xi(t), \quad t \geq 0, \\ \hat{\Delta}(t_0) = 0 \text{ or } \xi(t_0) = -\beta E(t_0), \end{cases}$$

不确定项补偿: $U(t) = -\bar{K}_p E(t) - \bar{K}_i Z(t) - \bar{B}^{-1} \hat{\Delta}(t)$



定理. 对于任意给定的不确定系统集合 $\mathcal{B}_{\bar{B}}^q \times \mathcal{D}_{h=0}^d \times \mathcal{F}_{g(\rho)} \times \Omega_{\rho_0} \times \Omega_{\rho^*}$ 当自抗扰控制器参数满足

$$\beta > \frac{2\rho_2 + \frac{1}{\underline{c}_i^2} + \rho_2^2(\bar{c}_i^2 + \bar{c}_p^2) + q^2\bar{c}_i^2 + \frac{(\underline{c}_i+1)^2}{\underline{c}_i^2\bar{c}_p^2}}{2(1-q)},$$

则有 $\begin{cases} \sup_{t \in [t_0, \infty)} \|E(t)\| \leq \rho_1, \\ \lim_{t \rightarrow \infty} \|E(t)\| = 0, \end{cases}$

$$\begin{cases} \rho_1 = \sqrt{\frac{1}{\underline{c}_J} (g(\rho_0)\rho_0 + q\bar{c}_p\rho_0^*)^2 + \frac{\bar{c}_J}{\underline{c}_J} \left(\rho_0^{*2} + \frac{1}{\underline{c}_i^2} d^2\right)}, \\ \rho_0^* = \rho_0 + \rho^*, \quad \rho_2 = g(\rho_1 + \rho^*) + \bar{c}_p q. \end{cases}$$

**针对PI控制系统的
模块化自抗扰调参理论**

W. Xue, etc., On Integrating Uncertainty Estimator into PI Control for A Class of Nonlinear Uncertain Systems, IEEE Transactions on Automatic Control, 2020.

自抗扰控制的思想与基本结构

- 自抗扰控制的思想
 - 自抗扰控制的非线性设计方法
 - 自抗扰控制的线性设计方法
 - 自抗扰控制与PID控制的关系
 - 自抗扰控制的一些应用实例
-

ADRC in U.S.: Milestones

- 1997: made the 1st successful ADRC hardware test on a servo mechanism
- 2008: \$1M venture capital, grew by \$5M in 2012.
- 2010: 1st factory implementation, 10 Parker extrusion lines (挤压机生产线) at a Parker Hannifin Extrusion Plant in North America (cpk: from 2.3 to >8; avg. 节能 57%)



- 2011: implemented ADRC in several high energy particle accelerators (高能粒子加速器) in the National Superconducting Cyclotron (超导回旋加速器) Lab in the U.S.



- 2011: Texas Instrument adopts ADRC; 3 patents granted.
- 2013: Texas Instrument released the ADRC based motion control chips (德州仪器的运动控制芯片)

从学术到工业的跨越

ADRC as a powerful solution has been explored in many domain of control engineering:

- **电机控制** (speed control of induction motor and permanent-magnet synchronous, noncircular turning process, etc)
- **飞行控制** (attitude control of space ship, high-speed vehicles, UAV, morphing aircraft, etc)
- **机器人** (force control, uncalibrated hand-eye coordination, AUV, etc)
- **热工过程** (unstable heat conduction systems, boiler-turbine-generator systems, ALSTOM gasifier, fractional-order system, etc)
- **电力电子装置** (DC-to DC power converters, rectifiers, inverters, HVDC SMC systems, etc)
- **飞行器**
- **电力系统、发动机、汽车、内燃机 ……**

2018年：珠海航展进行过失速机动



长征八号来了！从它身上能看到未来火箭的影子

科技日报 1周前

◎ 王伟童 高崇芮 董佳莹 科技日报记者 付毅飞

2020年12月22日12时37分，由中国航天科技集团一院抓总研制的我国新一代中型运载火箭长征八号，从文昌航天发射场点火升空，将新技术验证七号、海丝一号、元光号、天启星座零八星、智星一号A星5颗卫星送入预定轨道，圆满完成首次飞行任务。



+

i/s/8DVNOcND7t3ppXM2gcReTA



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多项新技术让火箭更聪明

近年来，我国运载火箭高密度发射已呈常态化，但也偶尔发生飞行故障甚至飞行失利的案例。如果火箭具备故障诊断和自主飞行能力，在发生故障时，自主调整，飞行结果便能大幅改善。

科研人员在长征八号身上，通过研究分析各种减载稳定控制方法，并采用自抗扰技术进行实时补偿控制，提高主动减载的效果，解决了大整流罩带来的静不稳定性大的难题。

同时，基于控制效果的喷管极性辨识和控制重构算法，能让运载火箭更聪明。通过技术的应用，长征八号具备了滑行段飞行故障在线辨识能力，能够在特定故障工况下自主进行姿态控制重构，提升了飞行控制适应性和智能化水平。

针对当前我国运载火箭测试周期长、在发射区占位时间长的的问题，科研团队积极探索火箭快速发射路径，为长征八号设计了“两平一垂”发射模式，即水平组装、水平状态整体运输、星罩组合体垂直转场对接。**预计在2022年左右，长征八号将实现“两平一垂”发射。届时发射区将不再需要规模庞大、组成复杂的塔架，可减少建设成本。**



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关注该公众号

Yi Huang, plenary talk of the 33rd Chinese Control Conference, 2014.



Outline

- Principles and methods of ADRC
- Practice
- Analysis
- Conclusions



brilliant idea:

**treating the unknown model as a special state (named extended state by Han)
using a special state observer, the extended state observer (ESO), to estimate it
in real time**

In such a brave and elegant stroke:

**a system, which may be uncertain, nonlinear, and time varying $\Rightarrow$
reduced to the canonical form of the linear chain of integrators**

<http://tcct.amss.ac.cn/newsletter/2014/201408/#plenarytalk>

Zhiqiang Gao, Cleveland State University, U.S.A

space.bilibili.com/408884199

自抗扰控制-ADRC

QQ: 914841616 ResearchGate: Zhiqiang_Gao5; Weixin: ADRC_ERA

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代表作

ADRC with disturbance: no oscillation 01:43

2017 Two-mass Positioning

3690 4

韩京清 (1937-2008)

自抗扰控制及其应用

25:09

1.4万 4

控制理论——模型论还是控制论¹⁾

韩京清

01:39:14

9117 51

公告

自抗扰控制作为控制器设计的基本原理, 1) 继承了PID的优势又突破了其结构和调参限制; 2) 吸收了现代控制论的精华又突破了其模型限制; 3) 为控制技术和科学的发展提供了一个新的支撑点, 让我们重新审视控制理论的前提和基本理念, 为它的重构和发展提供借鉴。

个人资料

UID 408884199 生日 01-01

自抗扰控制

TA的视频

129 最新发布 最多播放 最多收藏

播放全部 更多

- Texas Instrument,
- Freescale,
- Danfoss,
- Rockwell,
- Parker,
- Power plant in china,
- ...



2013 年, 美国德州仪器公司推出一系列基于自抗扰控制算法的运动控制芯片



2017 年 ADRC 在山西同达电厂 300MW 循环流化床机组协调控制回路投运



ADRC 用于多用途服务机器人的运动规划和避障控制中

谢谢！

薛文超

中国科学院数学与系统科学研究院 系统科学研究所

wenchaoxue@amss.ac.cn
